Web Dev Mastery

🌐 CSS Media Queries – Responsive Design

✅ What is Responsive Design?

Responsive design means creating web pages that look good on **all screen sizes**, from **mobile phones to large desktops**.

🔁 What are Media Queries?

Media Queries allow you to apply CSS styles depending on the **device's screen size**, **resolution**, **orientation**, and **more**.

📌 Syntax:

css

```
@media only screen and (max-width: 768px) {
  /* CSS rules for screens up to 768px wide */
}
```

- `only screen`: targets screens (not print)
- `max-width`: applies if screen width is **less than or equal to** given value
- `min-width`: applies if screen width is **greater than or equal to** given value

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📱 Common Breakpoints (Industry Standard)

Device Type	Breakpoint Range	Media Query Example
Extra Small (Mobile)	0px – 575px	`@media (max-width: 575px)`
Small (Mobile)	576px – 767px	`@media (min-width: 576px)`
Medium (Tablet)	768px – 991px	`@media (min-width: 768px)`
Large (Laptop)	992px – 1199px	`@media (min-width: 992px)`
Extra Large (Desktop)	1200px and up	`@media (min-width: 1200px)`

📊 Explanation of Breakpoints with Examples

1. 📱 Extra Small Devices (0 - 575px)

Use for small phones (iPhone SE, Android small devices)

css

```
@media (max-width: 575px) {
  body {
    background: lightblue;
    font-size: 14px;
  }
}
```

---

2. 📱 Small Devices (576px - 767px)

Use for standard smartphones

CSS
```css
@media (min-width: 576px) and (max-width: 767px) {
  .container {
    padding: 10px;
    flex-direction: column;
  }
}
```

---

3. 📲 Medium Devices (768px - 991px)

Use for tablets (iPad, Tab)

CSS
```css
@media (min-width: 768px) and (max-width: 991px) {
  .menu {
    display: none;
  }
}
```

---

4. 💻 Large Devices (992px - 1199px)

Use for laptops or desktops

CSS
```css
@media (min-width: 992px) and (max-width: 1199px) {
  .sidebar {
    width: 250px;
  }
}
```

---

5. 🖥️ Extra Large Devices (1200px and up)

Use for large desktops and TVs

css
```
@media (min-width: 1200px) {
  .container {
    max-width: 1140px;
    margin: auto;
  }
}
```

⚠️ Tips to Remember

- Use `max-width` when writing **mobile-first** CSS (recommended).
- Combine media queries with **flexbox** or **grid** for best layout.
- Always test your design on different screen sizes using browser dev tools.

🧪 Mini Project: Responsive Card Layout

🗒️ HTML

html
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
  <title>Responsive Cards</title>
  <link rel="stylesheet" href="style.css" />
</head>
<body>
  <h1>Responsive Card Layout</h1>
```

```
  <div class="card-container">
    <div class="card">Card 1</div>
    <div class="card">Card 2</div>
    <div class="card">Card 3</div>
  </div>
</body>
</html>
```

🎨 CSS (style.css)

css

```
body {
  font-family: sans-serif;
  margin: 0;
  padding: 20px;
  text-align: center;
}

.card-container {
  display: flex;
  justify-content: center;
  gap: 20px;
  flex-wrap: wrap;
}

.card {
  background: #f2f2f2;
  padding: 30px;
  width: 300px;
  border-radius: 8px;
  box-shadow: 0 0 10px #ccc;
}

/* Small Devices */
@media (max-width: 575px) {
  .card {
    width: 100%;
  }
```

```
}

/* Tablets */
@media (min-width: 576px) and (max-width: 991px) {
  .card {
    width: 45%;
  }
}

/* Laptops & Desktops */
@media (min-width: 992px) {
  .card {
    width: 30%;
  }
}
```

✅ Output Behavior:

- 📱 On mobile: Cards will stack full width.
- 📲 On tablet: Two cards in a row.
- 💻 On desktop: Three cards side by side.

🧠 Summary

- Media Queries help create flexible, mobile-friendly designs.
- Use min-width (mobile-first) approach for best practices.
- Test responsiveness using browser DevTools (F12 > Responsive Mode).
- Use percentages, `flexbox`, and `grid` for better layouts.